

Figure 6: A rotation transformation

Practice Problems

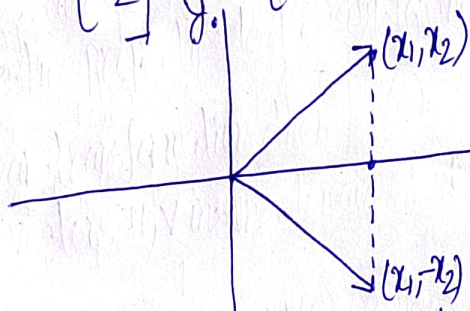
1. Suppose $T: \mathbb{R}^5 \rightarrow \mathbb{R}^2$ and $T(x) = Ax$ for some matrix A and for each x in $\mathbb{R}^5$. How many rows and columns does A have?

Solution: x must be a 5×1 matrix. Ax must be a 2×1 vector. By defn, Ax is the set of all linear combinations of the columns of A . $\therefore$ The matrix A must have 2 rows. It should have 5 columns. $\square$

2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Give a geometric description of the transformation $x \mapsto Ax$.

Solution: Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. $Ax = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \cdot x_1 + 0 \cdot x_2 \\ 0 \cdot x_1 + (-1) \cdot x_2 \end{bmatrix}$ (By Row-Vector Rule)

$$= \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}$$



The transformation matrix A reflects any arbitrary point about the x -axis. $\square$

3. The line segment from 0 to a vector u is the set of points of the form tu , where $0 \leq t \leq 1$. Show that a linear transformation T maps this segment into the segment between 0 and $T(u)$.

Solution: Let x be an arbitrary point on the line segment from 0 to a vector u . $\therefore x = tu$. $T(x) = T(tu) = tT(u)$ (by defn. of Linear Transformation condition 2). So, T maps x to a point $tT(u)$ on the segment between 0 and $T(u)$. $\square$